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Vertical channel semiconductor device with ferroelectric-insulator gate stack

Abstract

A semiconductor device includes a plurality of first conductive lines extending in a first direction and spaced apart from each other in a second direction intersecting the first direction, the first direction and second direction being horizontal directions, a plurality of vertical semiconductor patterns disposed on the plurality of first conductive lines, respectively, a gate electrode crossing the plurality of first conductive lines and penetrating each of the plurality of vertical semiconductor patterns, a ferroelectric pattern between the gate electrode and each of the plurality of vertical semiconductor patterns, and a gate insulating pattern between the ferroelectric pattern and each of the plurality of vertical semiconductor patterns.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This U.S. non-provisional patent application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0079264, filed on Jun. 18, 2021, in the Korean Intellectual Property Office, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

(2) Embodiments of the inventive concepts relate to semiconductor devices and methods of manufacturing the same, and more particularly, to semiconductor memory devices including vertical channel transistors and methods of manufacturing the same.

(3) Semiconductor memory devices may be classified into volatile memory devices and non-volatile memory devices. The volatile memory devices may lose their stored data when their power supplies are interrupted. For example, the volatile memory devices may include dynamic random access memory (DRAM) devices and static random access memory (SRAM) devices. On the contrary, the non-volatile memory devices may retain their stored data even when their power supplies are interrupted. For example, the non-volatile memory devices may include programmable ROM (PROM), erasable PROM (EPROM), electrically EPROM (EEPROM), and a flash memory device. In addition, next-generation non-volatile semiconductor memory devices (e.g., magnetic random access memory (MRAM) devices, phase-change random access memory (PRAM) devices,

and ferroelectric random access memory (FeRAM) devices) have been developed to provide high-performance and low power consumption semiconductor memory devices.
(4) Various techniques using semiconductor devices having different properties are being studied to improve an integration density and performance of a semiconductor device.
SUMMARY
(5) Embodiments of the inventive concepts may provide semiconductor devices capable of easily increasing an integration density and methods of manufacturing the same.
(6) Embodiments of the inventive concepts may also provide semiconductor devices capable of improving operating characteristics and reliability and methods of manufacturing the same.
(7) In an aspect, a semiconductor device may include a plurality of first conductive lines extending in a first direction and spaced apart from each other in a second direction intersecting the first direction, the first direction and second direction being horizontal directions, a plurality of vertical semiconductor patterns disposed on the plurality of first conductive lines, respectively, a gate electrode crossing the plurality of first conductive lines and penetrating each of the plurality of vertical semiconductor patterns, a ferroelectric pattern between the gate electrode and each of the plurality of vertical semiconductor patterns, and a gate insulating pattern between the ferroelectric pattern and each of the plurality of vertical semiconductor patterns.
(8) In an aspect, a semiconductor device may include a plurality of vertical semiconductor patterns on a substrate, the plurality of vertical semiconductor patterns spaced apart from each other in a first direction and a second direction which are parallel to a top surface of the substrate and intersect each other, the plurality of vertical semiconductor patterns extending in a third direction perpendicular to the top surface of the substrate, and a plurality of gate structures spaced apart from each other in the first direction and extending in the second direction on the substrate. Each of the plurality of gate structures may penetrate each vertical semiconductor pattern of corresponding vertical semiconductor patterns, spaced apart from each other in the second direction, of the plurality of vertical semiconductor patterns. Each of the plurality of gate structures may include a gate electrode penetrating the corresponding vertical semiconductor patterns, a ferroelectric pattern between the gate electrode and each of the corresponding vertical semiconductor patterns, and a gate insulating pattern between the ferroelectric pattern and each of the corresponding vertical semiconductor patterns.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The inventive concepts will become more apparent in view of the attached drawings and accompanying detailed description.
(2) FIG. **1** is a schematic perspective view illustrating a semiconductor device according to some embodiments of the inventive concepts.
(3) FIG. **2** is a plan view illustrating a semiconductor device according to some embodiments of the inventive concepts.
(4) FIG. **3** is a cross-sectional view taken along lines A-A′ and B-B′ of FIG. **2**.
(5) FIGS. **4**A and **4**B are enlarged views of a portion ‘PP**1**’ of FIG. **3**.
(6) FIG. **5** is a graph showing threshold voltage properties of a ferroelectric field-effect transistor according to some embodiments of the inventive concepts.
(7) FIGS. **6**, **8**, **10**, **12**, **14**, **16**, **18** and **20** are plan views illustrating a method of manufacturing a semiconductor device according to some embodiments of the inventive concepts.
(8) FIGS. **7**, **9**, **11**, **13**, **15**, **17**, **19** and **21** are cross-sectional views taken along lines A-A′ and B-B′ of FIGS. **6**, **8**, **10**, **12**, **14**, **16**, **18** and **20**, respectively.
(9) FIG. **22** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate

a semiconductor device according to some embodiments of the inventive concepts.

(10) FIG. **23** is an enlarged view of a portion 'PP**2**' of FIG. **22**.

(11) FIG. **24** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts.

(12) FIG. **25** is a plan view illustrating a semiconductor device according to some embodiments of the inventive concepts.

(13) FIG. **26** is a cross-sectional view taken along lines A-A′ and B-B′ of FIG. **25**.

(14) FIG. **27** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts.

(15) FIG. **28** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(16) Hereinafter, embodiments of the inventive concepts will be described in detail with reference to the accompanying drawings.

(17) FIG. **1** is a schematic perspective view illustrating a semiconductor device according to some embodiments of the inventive concepts.

(18) Referring to FIG. **1**, a plurality of first conductive lines **110** may extend in a first direction D**1** and may be spaced apart from each other in a second direction D**2** intersecting the first direction D**1** (e.g., perpendicular to the first direction D**1**). The plurality of first conductive lines **110** may be formed in or on a semiconductor substrate **100**. A plurality of vertical semiconductor patterns VC may be disposed on the plurality of first conductive lines **110**. The plurality of vertical semiconductor patterns VC may be two-dimensionally arranged (from a plan view) in the first direction D**1** and the second direction D**2** and may be spaced apart from each other in the first direction D**1** and the second direction D**2**. Vertical semiconductor patterns VC (e.g., a set of vertical semiconductor patterns VC), spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC may be connected to a corresponding one of the plurality of first conductive lines **110**. Vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC may be connected to the plurality of first conductive lines **110**, respectively. Each of the plurality of first conductive lines **110** may be connected to a set of corresponding vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC. The plurality of vertical semiconductor patterns VC may extend (e.g., lengthwise) in a third direction D**3** perpendicular to both the first direction D**1** and the second direction D**2**. The first direction D**1** and second direction D**2** may be described as horizontal directions, and the third direction D**3** may be described as a vertical direction. An item, layer, or portion of an item or layer described as extending "lengthwise" in a particular direction has a length in the particular direction and a width perpendicular to that direction, where the length is greater than the width.

(19) A plurality of gate structures GS may be disposed on the plurality of first conductive lines **110** and may cross the plurality of first conductive lines **110** (e.g., from a plan view). The plurality of gate structures GS may be spaced apart from each other in the first direction D**1** and may extend (e.g., lengthwise) in the second direction D**2**. Each of the plurality of gate structures GS may penetrate and pass through each vertical semiconductor pattern VA of a set of corresponding vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC.

(20) Each of the plurality of gate structures GS may include a gate electrode **130** extending in the second direction D**2**. The gate electrode **130** may penetrate the each vertical semiconductor pattern VC of the set of corresponding vertical semiconductor patterns VC and may cross the plurality of first conductive lines **110** (e.g., from a plan view). Each of the corresponding vertical semiconductor patterns VC may surround an outer surface of the gate electrode **130**. Each of the plurality of gate structures GS may further include a ferroelectric pattern **132** between the gate

electrode **130** and each of the corresponding vertical semiconductor patterns VC, and a gate insulating pattern **134** between the ferroelectric pattern **132** and each of the corresponding vertical semiconductor patterns VC. The ferroelectric pattern **132** may surround the outer surface of the gate electrode **130** and may extend (e.g., lengthwise) in the second direction D**2**. The gate insulating pattern **134** may surround an outer surface of the ferroelectric pattern **132** and may extend (e.g., lengthwise) in the second direction D**2**. Each of the corresponding vertical semiconductor patterns VC may surround an outer surface of the gate insulating pattern **134**. The gate electrode **130**, the ferroelectric pattern **132** and the gate insulating pattern **134** may be disposed inside each of the corresponding vertical semiconductor patterns VC, to be completely surrounded in the first direction D**1** and third direction D**3** by the corresponding vertical semiconductor patterns VC where they intersect the corresponding vertical semiconductor patterns VC.
(21) A plurality of second conductive lines **150** may be disposed on the plurality of vertical semiconductor patterns VC. The plurality of second conductive lines **150** may cross the plurality of gate structures GS (e.g., from a plan view). The plurality of second conductive lines **150** may extend (e.g., lengthwise) in the first direction D**1** and may be spaced apart from each other in the second direction D**2**. The vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC may be connected to a corresponding one of the plurality of second conductive lines **150**. The vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC may be connected to the plurality of second conductive lines **150**, respectively. Each of the plurality of second conductive lines **150** may be connected to corresponding vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC.
(22) FIG. **2** is a plan view illustrating a semiconductor device according to some embodiments of the inventive concepts, and FIG. **3** is a cross-sectional view taken along lines A-A′ and B-B′ of FIG. **2**. FIGS. **4**A and **4**B are enlarged views of a portion ‘PP**1**’ of FIG. **3**. FIG. **5** is a graph showing threshold voltage properties of a ferroelectric field-effect transistor according to some embodiments of the inventive concepts.
(23) Referring to FIGS. **2** and **3**, as well as FIG. **1**, a plurality of first conductive lines **110** may be disposed on or in a substrate **100**. The substrate **100** may include or may be a semiconductor substrate (e.g., a silicon (Si) substrate, a germanium (Ge) substrate, or a silicon-germanium (SiGe) substrate). The plurality of first conductive lines **110** may extend in the first direction D**1** and may be spaced apart from each other in the second direction D**2**. The first direction D**1** and the second direction D**2** may be parallel to a top surface **100**U of the substrate **100** and may intersect each other. The plurality of first conductive lines **110** may be buried in the substrate **100**. The plurality of first conductive lines **110** may be formed of or may include a conductive material and may be formed of or include, for example, doped poly-silicon, a metal, a conductive metal nitride, a conductive metal silicide, a conductive metal oxide, or any combination thereof. For example, the plurality of first conductive lines **110** may be formed of doped poly-silicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, or any combination thereof. However, embodiments of the inventive concepts are not limited thereto. In certain embodiments, the plurality of first conductive lines **110** may include or be formed of a two-dimensional semiconductor material. For example, the two-dimensional semiconductor material may include or be graphene, carbon nanotube, or a combination thereof.
(24) A buried insulating pattern **112** may be disposed between the substrate **100** and each of the plurality of first conductive lines **110**. The buried insulating pattern **112** may be disposed between the substrate **100** and a bottom surface of each of the plurality of first conductive lines **110** and may extend between the substrate **100** and sidewalls of each of the plurality of first conductive lines **110**. Each of the plurality of first conductive lines **110** may be spaced apart from the substrate **100** with the buried insulating pattern **112** interposed therebetween. The buried insulating pattern **112**

may include or be formed of an insulating material (e.g., silicon oxide, silicon nitride, and/or silicon oxynitride).

(25) A plurality of vertical semiconductor patterns VC may be disposed on the plurality of first conductive lines **110**. The plurality of vertical semiconductor patterns VC may be two-dimensionally arranged (from a plan view) in the first direction D**1** and the second direction D**2** and may be spaced apart from each other in the first direction D**1** and the second direction D**2**. The plurality of vertical semiconductor patterns VC may extend in the third direction D**3** perpendicular to the top surface **100**U of the substrate **100**. Vertical semiconductor patterns VC (e.g., a set of vertical semiconductor patterns VC), spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC may be connected to a corresponding one of the plurality of first conductive lines **110**. Vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC may be connected to the plurality of first conductive lines **110**, respectively. Each of the plurality of first conductive lines **110** may be connected to corresponding vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC.

(26) For example, the plurality of vertical semiconductor patterns VC may include or be formed of silicon, germanium, silicon-germanium, an oxide semiconductor, MoS.sub.2, or WS.sub.2. The oxide semiconductor may include or may be InGaZnO, InGaSiO, InSnZnO, InZnO, ZnO, ZnSnO, ZnON, ZrZnSnO, SnO, HfInZnO, GaZnSnO, AlZnSnO, YbGaZnO, InGaO, or any combination thereof. The plurality of vertical semiconductor patterns VC may include or be formed of, for example, poly-crystalline silicon. In certain embodiments, the plurality of vertical semiconductor patterns VC may include or be formed of a two-dimensional semiconductor material. For example, the two-dimensional semiconductor material may include or be graphene, carbon nanotube, or a combination thereof.

(27) Each of the plurality of vertical semiconductor patterns VC may include a first dopant region **122**A and a second dopant region **122**B spaced apart from each other in the third direction D**3**, and a channel region **120** between the first dopant region **122**A and the second dopant region **122**B. The first dopant region **122**A and the second dopant region **122**B may include dopants having the same conductivity type. For example, the first dopant region **122**A and the second dopant region **122**B may include N-type dopants or P-type dopants. The first dopant regions **122**A of the plurality of vertical semiconductor patterns VC may be adjacent to the plurality of first conductive lines **110** and may be connected to (e.g., directly connected to) the plurality of first conductive lines **110**. It will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element, there are no intervening elements present at the point of connection or contact.

(28) A plurality of gate structures GS may be disposed on the plurality of first conductive lines **110** and may cross the plurality of first conductive lines **110**. The plurality of gate structures GS may be spaced apart from each other in the first direction D**1** and may extend in the second direction D**2**. Each of the plurality of gate structures GS may penetrate corresponding vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC. Each of the plurality of gate structures GS may penetrate the channel regions **120** of the corresponding vertical semiconductor patterns VC. Each of the plurality of vertical semiconductor patterns VC may surround an outer surface GS_S of a corresponding gate structure GS of the plurality of gate structures GS. The channel region **120** of each of the plurality of vertical semiconductor patterns VC may surround the outer surface GS_S of the corresponding gate structure GS.

(29) Each of the plurality of gate structures GS may include a gate electrode **130** extending in the

second direction D**2** to penetrate the corresponding vertical semiconductor patterns VC, a ferroelectric pattern **132** between the gate electrode **130** and each of the corresponding vertical semiconductor patterns VC, and a gate insulating pattern **134** between the ferroelectric pattern **132** and each of the corresponding vertical semiconductor patterns VC. The gate electrode **130** may include or be formed of doped poly-silicon, a metal, a conductive metal nitride, a conductive metal silicide, a conductive metal oxide, or any combination thereof. For example, the gate electrode **130** may be formed of doped poly-silicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, or any combination thereof. However, embodiments of the inventive concepts are not limited thereto. The ferroelectric pattern **132** may include or be formed of a hafnium (Hf) compound having ferroelectric properties. For example, the ferroelectric pattern **132** may include or be formed of HfO.sub.2, HfZnO, HfSiO, HfSiON, HffaO, HfTiO, HfZrO, or any combination thereof. The ferroelectric pattern **132** may have an orthorhombic phase. The gate insulating pattern **134** may include or be formed of a silicon oxide layer, a silicon oxynitride layer, a high-k dielectric layer having a dielectric constant higher than that of a silicon oxide layer, or any combination thereof. The high-k dielectric layer may include or may be a metal oxide or a metal oxynitride.

(30) The gate electrode **130** may be disposed inside each of the corresponding vertical semiconductor patterns VC, and each of the corresponding vertical semiconductor patterns VC may surround an outer surface **130**S of the gate electrode **130**. The ferroelectric pattern **132** may be disposed between the outer surface **130**S of the gate electrode **130** and each of the corresponding vertical semiconductor patterns VC. The ferroelectric pattern **132** may surround the outer surface **130**S of the gate electrode **130** and may extend in the second direction D**2** along the gate electrode **130**. The ferroelectric pattern **132** may be disposed inside each of the corresponding vertical semiconductor patterns VC, and each of the corresponding vertical semiconductor patterns VC may surround an outer surface **132**S of the ferroelectric pattern **132**. The gate insulating pattern **134** may be disposed between the outer surface **132**S of the ferroelectric pattern **132** and each of the corresponding vertical semiconductor patterns VC. The gate insulating pattern **134** may surround the outer surface **132**S of the ferroelectric pattern **132** and may extend in the second direction D**2** along the gate electrode **130**. The gate insulating pattern **134** may be disposed inside each of the corresponding vertical semiconductor patterns VC, and each of the corresponding vertical semiconductor patterns VC may surround, and may contact, an outer surface **134**S of the gate insulating pattern **134**. The gate electrode **130**, the ferroelectric pattern **132** and the gate insulating pattern **134** may be disposed in the channel region **120** of each of the corresponding vertical semiconductor patterns VC, and the channel region **120** of each of the corresponding vertical semiconductor patterns VC may surround the outer surfaces **130**S, **132**S and **134**S of the gate electrode **130**, the ferroelectric pattern **132** and the gate insulating pattern **134**. The outer surface GS_S of each of the plurality of gate structures GS may correspond to the outer surface **134**S of the gate insulating pattern **134**.

(31) In some embodiments, the outer surface **130**S of the gate electrode **130** may have a rounded shape when viewed in a cross-sectional view. For example, the gate electrode **130** may have a cylindrical shape. Each of the ferroelectric pattern **132** and the gate insulating pattern **134** may have a rounded ring shape surrounding the outer surface **130**S of the gate electrode **130** when viewed in a cross-sectional view. The outer surface GS_S of each of the plurality of gate structures GS may have a rounded shape when viewed in a cross-sectional view. For example, each of the plurality of gate structures GS may have a cylindrical shape.

(32) Support patterns **160** may be disposed on the substrate **100** and may support the plurality of gate structures GS. The support patterns **160** may be spaced apart from each other in the second direction D**2** with the plurality of gate structures GS interposed therebetween and may extend in the first direction D**1**. One of the support patterns **160** may be in contact with end portions (e.g., first end portions) of the plurality of gate structures GS, and another of the support patterns **160** (e.g., a

next adjacent support pattern **160**) may be in contact with other end portions (e.g., second end portions opposite the first end portions) of the plurality of gate structures GS. Each of the support patterns **160** may include a first insulating pattern **162**, a sacrificial pattern **164** and a second insulating pattern **166**, which are sequentially stacked on the substrate **100** in the third direction D**3**. The sacrificial pattern **164** may include or be formed of a material having an etch selectivity with respect to the first and second insulating patterns **162** and **166**. For example, the sacrificial pattern **164** may include or be formed of silicon nitride, and the first and second insulating patterns **162** and **166** may include or be formed of silicon oxide. The sacrificial pattern **164** of one of the support patterns **160** (e.g., a first support pattern **160**) may be in contact with first end portions of the plurality of gate structures GS, and the sacrificial pattern **164** of the other of the support patterns **160** (e.g., a second support pattern **160**) may be in contact with the second end portions of the plurality of gate structures GS opposite the first end portions. Ordinal numbers such as "first," "second," "third," etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using "first," "second," etc., in the specification, may still be referred to as "first" or "second" in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., "first" in a particular claim) may be described elsewhere with a different ordinal number (e.g., "second" in the specification or another claim).

(33) An interlayer insulating layer **140** may be disposed on the substrate **100** and may cover the plurality of first conductive lines **110**, the plurality of vertical semiconductor patterns VC, and the plurality of gate structures GS. The interlayer insulating layer **140** may fill a space between the substrate **100** and the plurality of gate structures GS and between the plurality of vertical semiconductor patterns VC. The interlayer insulating layer **140** may not cover top surfaces of the plurality of vertical semiconductor patterns VC, thereby leaving those top surfaces exposed with respect to the interlayer insulating layer **140**. For example, the interlayer insulating layer **140** may include or may be formed of at least one of silicon oxide, silicon nitride, or silicon oxynitride.

(34) A plurality of second conductive lines **150** may be disposed on the interlayer insulating layer **140** and the plurality of vertical semiconductor patterns VC. The plurality of second conductive lines **150** may cross the plurality of gate structures GS. The plurality of second conductive lines **150** may extend in the first direction D**1** and may be spaced apart from each other in the second direction D**2**. The vertical semiconductor patterns VC (e.g., a set of vertical semiconductor patterns VC), spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC may be connected to a corresponding one of the plurality of second conductive lines **150**. The vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC may be connected to the plurality of second conductive lines **150**, respectively. Each of the plurality of second conductive lines **150** may be connected to corresponding vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC. The second dopant regions **122**B of the plurality of vertical semiconductor patterns VC may be adjacent to the plurality of second conductive lines **150** and may be connected to and may contact the plurality of second conductive lines **150**. The plurality of second conductive lines **150** may include or be formed of a conductive material and may include or be formed of, for example, doped poly-silicon, a metal, a conductive metal nitride, a conductive metal silicide, a conductive metal oxide, or any combination thereof. For example, the plurality of second conductive lines **150** may be formed of doped poly-silicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, or any combination thereof. However, embodiments of the inventive concepts are not limited thereto. In certain embodiments, the plurality of second conductive lines **150** may include or be formed of a two-dimensional semiconductor material. For example, the two-dimensional semiconductor material may include or may be graphene, carbon nanotube, or a combination thereof.

(35) In some embodiments, the plurality of first conductive lines **110** may function as bit lines, and the plurality of second conductive lines **150** may function as source lines. In certain embodiments, the plurality of first conductive lines **110** may function as source lines, and the plurality of second conductive lines **150** may function as bit lines.
(36) Referring to FIGS. **4**A, **4**B and **5**, each of the plurality of vertical semiconductor patterns VC and a corresponding gate structure GS penetrating therethrough may constitute a ferroelectric field-effect transistor. Hereinafter, an operating principle of an N-type ferroelectric field-effect transistor will be described as an example for the purpose of ease and convenience in explanation. However, embodiments of the inventive concepts are not limited thereto. For example, referring to FIGS. **4**A and **5**, when a positive voltage is applied to the gate electrode **130** (V.sub.G>0), a polarity Pr of the ferroelectric pattern **132** may be in a positive (+) direction. Thus, a channel may be formed in the channel region **120**, and a current ID may flow between the first and second dopant regions **122**A and **122**B. In this case, the ferroelectric field-effect transistor may have a first threshold voltage V.sub.TH1 which is relatively small, and may be in an on-state. Even after the positive voltage applied to the gate electrode **130** is interrupted, the ferroelectric field-effect transistor may be maintained in the on-state due to a remaining polarity +Pr of the ferroelectric pattern **132**. Referring to FIGS. **4**B and **5**, when a negative voltage is applied to the gate electrode **130** (V.sub.G<0), the direction of the polarity Pr of the ferroelectric pattern **132** may be reversed to a negative (−) direction. In this case, the ferroelectric field-effect transistor may have a second threshold voltage V.sub.TH2 greater than the first threshold voltage Vim and may be in an off-state. Even though a voltage applied to the gate electrode **130** increases to 0V, the ferroelectric field-effect transistor may be maintained in the off-state due to a remaining reversed polarity −Pr of the ferroelectric pattern **132**. Thus, according to the embodiments of the inventive concepts, a non-volatile semiconductor memory device including the ferroelectric field-effect transistor may be provided.
(37) According to the embodiments of the inventive concepts, the plurality of vertical semiconductor patterns VC may be two-dimensionally arranged in the first direction D**1** and the second direction D**2** on the substrate **100**, and each of the plurality of gate structures GS may penetrate corresponding vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC. Each of the plurality of gate structures GS may be disposed inside the corresponding vertical semiconductor patterns VC, and thus it is possible to prevent disturbance between the gate structures GS adjacent to each other. In addition, since each of the plurality of gate structures GS is disposed inside the corresponding vertical semiconductor patterns VC, it may be easy to reduce a size of a memory cell structure including the plurality of gate structures GS and the plurality of vertical semiconductor patterns VC. Thus, an integration density of the semiconductor device may be easily improved.
(38) Furthermore, each of the plurality of gate structures GS may include the gate electrode **130**, the ferroelectric pattern **132** surrounding, and contacting, the outer surface **130**S of the gate electrode **130**, and the gate insulating pattern **134** surrounding, and contacting, the outer surface **132**S of the ferroelectric pattern **132**. Since the ferroelectric pattern **132** surrounds the outer surface **130**S of the gate electrode **130**, an intensity of an electric field applied to the ferroelectric pattern **132** may be increased, and an intensity of an electric field applied to the gate insulating pattern **134** may be reduced. Thus, polarity properties of the ferroelectric pattern **132** may be improved, and endurance of the gate insulating pattern **134** may be improved. As a result, operating characteristics and reliability of the semiconductor device may be improved.
(39) FIGS. **6**, **8**, **10**, **12**, **14**, **16**, **18** and **20** are plan views illustrating a method of manufacturing a semiconductor device according to some embodiments of the inventive concepts. FIGS. **7**, **9**, **11**, **13**, **15**, **17**, **19** and **21** are cross-sectional views taken along lines A-A′ and B-B′ of FIGS. **6**, **8**, **10**, **12**, **14**, **16**, **18** and **20**, respectively. Hereinafter, the descriptions to the same technical features as described with reference to FIGS. **1** to **5** will be omitted or mentioned briefly for the purpose of

ease and convenience in explanation.

(40) Referring to FIGS. **6** and **7**, a plurality of first conductive lines **110** may be formed in a substrate **100**, and a buried insulating pattern **112** may be formed between the substrate **100** and each of the plurality of first conductive lines **110**. The plurality of first conductive lines **110** may extend in the first direction D**1** and may be spaced apart from each other in the second direction D**2**. For example, the formation of the plurality of first conductive lines **110** and the buried insulating pattern **112** may include forming a plurality of trenches T by etching an upper portion of the substrate **100**, forming a buried insulating layer filling a portion of each of the plurality of trenches T on the substrate **100**, forming a first conductive layer filling a remaining portion of each of the plurality of trenches T on the buried insulating layer, and planarizing the first conductive layer and the buried insulating layer until a top surface **100**U of the substrate **100** is exposed. Each of the plurality of trenches T may have a line shape (e.g., straight line shape) extending in the first direction D**1**, and the plurality of trenches T may be spaced apart from each other in the second direction D**2**. The plurality of first conductive lines **110** and the buried insulating patterns **112** may be locally formed in the plurality of trenches T by the planarization of the first conductive layer and the buried insulating layer.

(41) A stack layer SS may be formed on the substrate **100** and may cover the plurality of first conductive lines **110** and the buried insulating patterns **112**. The stack layer SS may include a first insulating layer **162**L, a sacrificial layer **164**L and a second insulating layer **166**L, which are sequentially stacked in the third direction D**3**. The sacrificial layer **164**L may include a material having an etch selectivity with respect to the first and second insulating layers **162**L and **166**L. For example, the first and second insulating layers **162**L and **166**L may include or be formed of silicon oxide, and the sacrificial layer **164**L may include or be formed of silicon nitride.

(42) Referring to FIGS. **8** and **9**, a plurality of openings **160**OP may be formed in the stack layer SS. Each of the plurality of openings **160**OP may have a line shape extending in the second direction D**2**, and the plurality of openings **160**OP may be spaced apart from each other in the first direction D**1**. Each of the plurality of openings **160**OP may cross the plurality of first conductive lines **110** and may expose portions of the plurality of first conductive lines **110**. For example, the formation of the plurality of openings **160**OP may include forming a first mask pattern defining the plurality of openings **160**OP on the stack layer SS, sequentially etching the second insulating layer **166**L, the sacrificial layer **164**L and the first insulating layer **162**L by using the first mask pattern as an etch mask, and removing the first mask pattern.

(43) Since the plurality of openings **160**OP is formed in the stack layer SS, line patterns LP may be formed between the plurality of openings **160**OP, and support patterns **160** may be formed at both sides of each of the plurality of openings **160**OP, respectively. The line patterns LP may extend in the second direction D**2** and may be spaced apart from each other in the first direction D**1**. The support patterns **160** may be spaced apart from each other in the second direction D**2** with the line patterns LP interposed therebetween and may extend in the first direction D**1**. One of the support patterns **160** may be connected to end portions of the line patterns LP, and the other of the support patterns **160** may be connected to other end portions of the line patterns LP. The support patterns **160** and the line patterns LP may be connected to each other to constitute one body (e.g., one continuous, integrated structure). Each of the support patterns **160** and the line patterns LP may include a first insulating pattern **162**, a sacrificial pattern **164** and a second insulating pattern **166**, which are sequentially stacked on the substrate **100** in the third direction D**3**. The first insulating pattern **162**, the sacrificial pattern **164** and the second insulating pattern **166** may be formed by etching the first insulating layer **162**L, the sacrificial layer **164**L and the second insulating layer **166**L, respectively.

(44) Referring to FIGS. **10** and **11**, the sacrificial pattern **164** of each of the line patterns LP may be removed, and thus an empty region ER may be formed in each of the line patterns LP. The empty region ER may be formed between the first insulating pattern **162** and the second insulating pattern

166 of each of the line patterns LP. The empty region ER may have a line shape extending in the second direction D**2**. The empty regions ER in the line patterns LP may be spaced apart from each other in the first direction D**1**. The removal of the sacrificial pattern **164** of each of the line patterns LP may include performing a selective etching process (e.g., a wet etching process) having an etch selectivity with respect to the sacrificial pattern **164**. The first insulating pattern **162** and the second insulating pattern **166** of each of the line patterns LP may not be removed by the selective etching process.
(45) During the removal of the sacrificial pattern **164** of each of the line patterns LP, the sacrificial pattern **164** of each of the support patterns **160** may not be completely removed but may remain. The sacrificial pattern **164** of the one of the support patterns **160** may extend in the first direction D**1** and may be exposed by end portions of the empty regions ER of the line patterns LP. The sacrificial pattern **164** of the other of the support patterns **160** may be exposed by other end portions of the empty regions ER of the line patterns LP. The first insulating pattern **162** and the second insulating pattern **166** of each of the support patterns **160** may not be removed by the selective etching process.
(46) Referring to FIGS. **12** and **13**, a gate electrode **130** may be formed in the empty region ER of each of the line patterns LP. The gate electrode **130** may have a line shape extending in the second direction D**2**, and the gate electrodes **130** in the line patterns LP may be spaced apart from each other in the first direction D**1**. For example, the formation of the gate electrode **130** may include forming a gate electrode layer filling the empty region ER of each of the line patterns LP and filling at least a portion of each of the plurality of openings **160**OP, and removing a remaining portion of the gate electrode layer in the plurality of openings **160**OP. The remaining portion of the gate electrode layer in the plurality of openings **160**OP may be removed by, for example, an anisotropic etching process. The gate electrode **130** may be locally formed in the empty region ER by the removal of the remaining portion of the gate electrode layer in the plurality of openings **160**OP.
(47) The support patterns **160** may be spaced apart from each other in the second direction D**2** with the gate electrode **130** interposed therebetween. The sacrificial pattern **164** of the one of the support patterns **160** may be in contact with an end portion of the gate electrode **130**, and the sacrificial pattern **164** of the other of the support patterns **160** may be in contact with another end portion of the gate electrode **130**. The support patterns **160** may support the gate electrode **130** in subsequent processes.
(48) Referring to FIGS. **14** and **15**, the first insulating pattern **162** and the second insulating pattern **166** of each of the line patterns LP may be removed. For example, the removal of the first insulating pattern **162** and the second insulating pattern **166** of each of the line patterns LP may include forming a second mask pattern covering the support patterns **160** but exposing the line patterns LP, etching (dry or wet-etching) the first insulating pattern **162** and the second insulating pattern **166** of each of the line patterns LP by using the second mask pattern as an etch mask, and removing the second mask pattern. During the removal of the first insulating pattern **162** and the second insulating pattern **166** of each of the line patterns LP, the support patterns **160** support the gate electrode **130**.
(49) An outer surface **130**S of the gate electrode **130** may be exposed by the removal of the first insulating pattern **162** and the second insulating pattern **166** of each of the line patterns LP. In some embodiments, the outer surface **130**S of the gate electrode **130** may have a rounded shape when viewed in a cross-sectional view. The rounded shape of the outer surface **130**S of the gate electrode **130** may be formed by performing an additional etching process (i.e., an additional dry or wet etching process) on the outer surface **130**S of the gate electrode **130**. In certain embodiments, the additional etching process may be omitted. In this case, the outer surface **130**S of the gate electrode **130** may have an angular shape when viewed in a cross-sectional view. Thereafter, a ferroelectric pattern **132** may be formed to surround the outer surface **130**S of the gate electrode **130**, and a gate insulating pattern **134** may be formed to surround an outer surface **132**S of the ferroelectric pattern

132. For example, the ferroelectric pattern **132** and the gate insulating pattern **134** may be formed by selective deposition methods performed on the gate electrode **130**.
(50) The gate electrode **130**, the ferroelectric pattern **132** and the gate insulating pattern **134** may constitute a gate structure GS. The gate structure GS may have a line shape extending in the second direction D**2**, and a plurality of the gate structures GS may be spaced apart from each other in the first direction D**1**. The plurality of gate structures GS may cross the plurality of first conductive lines **110**. The support patterns **160** may be spaced apart from each other in the second direction D**2** with the plurality of gate structures GS interposed therebetween. One of the support patterns **160** may be in contact with end portions of the plurality of gate structures GS, and the other of the support patterns **160** may be in contact with other end portions of the plurality of gate structures GS. The support patterns **160** may support the plurality of gate structures GS.
(51) Referring to FIGS. **16** and **17**, a semiconductor layer **120**L may be formed on the substrate **100** to cover the plurality of gate structures GS. The semiconductor layer **120**L may surround an outer surface GS_S of each of the plurality of gate structures GS, and the outer surface GS_S of each of the plurality of gate structures GS may correspond to an outer surface **134**S of the gate insulating pattern **134**. The semiconductor layer **120**L may fill a space between the plurality of first conductive lines **110** and the plurality of gate structures GS, between the substrate **100** and the plurality of gate structures GS and between the support patterns **160**. The semiconductor layer **120**L may be formed by, for example, a chemical vapor deposition (CVD) method.
(52) In the formation of the semiconductor layer **120**L, a first dopant region **122**A may be formed in a lower portion of the semiconductor layer **120**L, and a second dopant region **122**B may be formed in an upper portion of the semiconductor layer **120**L. The first dopant region **122**A may be adjacent to (e.g., directly adjacent to) the substrate **100** and the plurality of first conductive lines **110**. For example, the formation of the first and second dopant regions **122**A and **122**B may include injecting dopants into the semiconductor layer **120**L in the formation of the semiconductor layer **120**L.
(53) Referring to FIGS. **18** and **19**, preliminary semiconductor patterns **120**P may be formed to intersect the plurality of first conductive lines **110**. The preliminary semiconductor patterns **120**P may extend in the second direction D**2** and may be spaced apart from each other in the first direction D**1**. The preliminary semiconductor patterns **120**P may overlap with the gate structures GS, respectively. The gate structures GS may be disposed in the preliminary semiconductor patterns **120**P, respectively, and each of the preliminary semiconductor patterns **120**P may surround the outer surface GS_S of each of the plurality of gate structures GS. The preliminary semiconductor patterns **120**P may be formed by patterning the semiconductor layer **120**L. The first dopant region **122**A may be disposed in a lower portion of each of the preliminary semiconductor patterns **120**P, and the second dopant region **122**B may be disposed in an upper portion of each of the preliminary semiconductor patterns **120**P.
(54) A first interlayer insulating layer **140**A may be formed to fill a space between the preliminary semiconductor patterns **120**P. The first interlayer insulating layer **140**A may cover sidewalls of the preliminary semiconductor patterns **120**P and may expose top surfaces of the preliminary semiconductor patterns **120**P. For example, the first interlayer insulating layer **140**A may be or may include at least one of silicon oxide, silicon nitride, or silicon oxynitride.
(55) Referring to FIGS. **20** and **21**, a plurality of vertical semiconductor patterns VC may be formed by patterning the preliminary semiconductor patterns **120**P. The plurality of vertical semiconductor patterns VC may be spaced apart from each other in the first direction D**1** and the second direction D**2**. Each of the preliminary semiconductor patterns **120**P may be etched by a dry or wet etching process and thus may be divided into the vertical semiconductor patterns VC spaced apart from each other in the second direction D**2**. Each of the plurality of vertical semiconductor patterns VC may include the first dopant region **122**A and the second dopant region **122**B spaced apart from each other in the third direction D**3**, and a channel region **120** between the first dopant

region **122**A and the second dopant region **122**B. A second interlayer insulating layer **140**B may be formed to fill a space between the vertical semiconductor patterns VC spaced apart from each other in the second direction D**2**. For example, the second interlayer insulating layer **140**B may include or may be formed of at least one of silicon oxide, silicon nitride, or silicon oxynitride. The first interlayer insulating layer **140**A and the second interlayer insulating layer **140**B may be referred to as an interlayer insulating layer **140**. The interlayer insulating layer **140** may not cover top surfaces of the plurality of vertical semiconductor patterns VC, and thus the top surfaces of the plurality of vertical semiconductor patterns VC may be exposed with respect to the interlayer insulating layer **140**.

(56) Vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC may be connected to a corresponding one of the plurality of first conductive lines **110**. Vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC may be connected to the plurality of first conductive lines **110**, respectively. Each of the plurality of gate structures GS may penetrate the vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC. Each of the plurality of vertical semiconductor patterns VC may surround the outer surface GS_S of a corresponding gate structure GS of the plurality of gate structures GS. The channel region **120** of each of the plurality of vertical semiconductor patterns VC may surround the outer surface GS_S of the corresponding gate structure GS.

(57) Referring again to FIGS. **2** and **3**, a plurality of second conductive lines **150** may be formed on the interlayer insulating layer **140** and the plurality of vertical semiconductor patterns VC. The plurality of second conductive lines **150** may be formed to cross the plurality of gate structures GS. The plurality of second conductive lines **150** may extend in the first direction D**1** and may be spaced apart from each other in the second direction D**2**. For example, the formation of the plurality of second conductive lines **150** may include forming a second conductive layer on the interlayer insulating layer **140**, and patterning the second conductive layer. The vertical semiconductor patterns VC, spaced apart from each other in the first direction D**1**, of the plurality of vertical semiconductor patterns VC may be connected to a corresponding one of the plurality of second conductive lines **150**. The vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC may be connected to the plurality of second conductive lines **150**, respectively.

(58) FIG. **22** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts. FIG. **23** is an enlarged view of a portion ‘PP**2**’ of FIG. **22**. Hereinafter, differences between the present embodiments and the embodiments described with reference to FIGS. **1** to **5** will be mainly described for the purpose of ease and convenience in explanation.

(59) Referring to FIGS. **22** and **23**, according to some embodiments, each of the plurality of gate structures GS may further include a metal pattern **136** between the ferroelectric pattern **132** and the gate insulating pattern **134**. The metal pattern **136** may include or may be formed of a metal (e.g., Pt, etc.) and/or a metal oxide (e.g., RuO.sub.2, IrO.sub.2, LaSrCoO.sub.3, etc.). The metal pattern **136** may be used to easily maintain the polarity Pr of the ferroelectric pattern **132** described with reference to FIGS. **4**A, **4**B and **5**.

(60) The metal pattern **136** may surround the outer surface **132**S of the ferroelectric pattern **132** and may extend in the second direction D**2** along the gate electrode **130**. The metal pattern **136** may be disposed inside each of corresponding vertical semiconductor patterns VC spaced apart from each other in the second direction D**2**, and each of the corresponding vertical semiconductor patterns VC may surround an outer surface **136**S of the metal pattern **136**. The gate insulating pattern **134** may be disposed between the outer surface **136**S of the metal pattern **136** and each of the corresponding vertical semiconductor patterns VC. The gate insulating pattern **134** may surround the outer surface

136S of the metal pattern **136** and may extend in the second direction D**2** along the gate electrode **130**. The gate insulating pattern **134** may be disposed inside each of the corresponding vertical semiconductor patterns VC, and each of the corresponding vertical semiconductor patterns VC may surround the outer surface **134**S of the gate insulating pattern **134**.
(61) The gate electrode **130**, the ferroelectric pattern **132**, the metal pattern **136** and the gate insulating pattern **134** may be disposed in the channel region **120** of each of the corresponding vertical semiconductor patterns VC, and the channel region **120** of each of the corresponding vertical semiconductor patterns VC may surround the outer surfaces **130**S, **132**S, **136**S and **134**S of the gate electrode **130**, the ferroelectric pattern **132**, the metal pattern **136** and the gate insulating pattern **134**.
(62) FIG. **24** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the embodiments described with reference to FIGS. **1** to **5** will be mainly described for the purpose of ease and convenience in explanation.
(63) Referring to FIG. **24**, according to some embodiments, the outer surface **130**S of the gate electrode **130** may have an angular shape when viewed in a cross-sectional view. For example, the gate electrode **130** may have an angular pillar shape. Each of the ferroelectric pattern **132** and the gate insulating pattern **134** may have an angular ring shape surrounding the outer surface **130**S of the gate electrode **130** when viewed in a cross-sectional view. The outer surface GS_S of each of the plurality of gate structures GS may have an angular shape when viewed in a cross-sectional view. For example, each of the plurality of gate structures GS may have an angular pillar shape. The angular shapes described for this embodiment may be a rectangular shape, For example.
(64) FIG. **25** is a plan view illustrating a semiconductor device according to some embodiments of the inventive concepts, and FIG. **26** is a cross-sectional view taken along lines A-A′ and B-B′ of FIG. **25**. Hereinafter, differences between the present embodiments and the embodiments described with reference to FIGS. **1** to **5** will be mainly described for the purpose of ease and convenience in explanation.
(65) Referring to FIGS. **25** and **26**, according to some embodiments, a common conductive line **150**C may be disposed on the interlayer insulating layer **140** and the plurality of vertical semiconductor patterns VC. The common conductive line **150**C may have a plate shape and may cover the plurality of vertical semiconductor patterns VC. The common conductive line **150**C may be connected in common to the plurality of vertical semiconductor patterns VC spaced apart from each other in the first direction D**1** and the second direction D**2**. In some embodiments, the plurality of first conductive lines **110** may function as bit lines, and the common conductive line **150**C may function as a common source line.
(66) FIG. **27** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the embodiments described with reference to FIGS. **1** to **5** will be mainly described for the purpose of ease and convenience in explanation. Furthermore, though the example embodiment of FIG. **2** is shown as part of FIG. **27**, the embodiments of the other figures (e.g., FIG. **22-26** or **28**) may be used in place of the embodiment of FIG. **2**. It should be noted that the different variations shown in FIGS. **22-28** can be combined in different ways so that the different features can be adjusted according to a desired effect.
(67) Referring to FIG. **27**, according to some embodiments, a peripheral circuit structure PTR, CT and CL may be disposed on the substrate **100**, and a lower interlayer insulating layer **180** may be disposed on the substrate **100** to cover the peripheral circuit structure PTR, CT and CL. For example, the lower interlayer insulating layer **180** may include or be formed of at least one of silicon oxide, silicon nitride, or silicon oxynitride.

(68) A cell structure including the plurality of first conductive lines **110**, the plurality of vertical semiconductor patterns VC, the plurality of gate structures GS and the plurality of second conductive lines **150** may be disposed on the lower interlayer insulating layer **180**. According to some embodiments, an upper layer **200** may be disposed on the lower interlayer insulating layer **180**, and the cell structure may be disposed on the upper layer **200**. For example, the upper layer **200** may be an insulating layer or a semiconductor layer. In some embodiments, the plurality of first conductive lines **110** may be buried in the upper layer **200**, and a buried insulating pattern **112** may be disposed between the upper layer **200** and each of the plurality of first conductive lines **110**.
(69) The peripheral circuit structure PTR, CT and CL may include peripheral transistors PTR on the substrate **100**. The peripheral transistors PTR may constitute a peripheral circuit for driving memory cells in the cell structure. The plurality of first conductive lines **110**, the plurality of gate structures GS and the plurality of second conductive lines **150** may be electrically connected to the peripheral transistors PTR and may be controlled by the peripheral transistors PTR. Each of the peripheral transistors PTR may include a peripheral gate electrode PGE on the substrate **100**, a peripheral gate insulating pattern PGI between the substrate **100** and the peripheral gate electrode PGE, and peripheral source/drain regions PSD disposed in the substrate **100** at both sides of the peripheral gate electrode PGE.
(70) The peripheral circuit structure PTR, CT and CL may further include peripheral contacts CT connected to terminals of the peripheral transistors PTR, and peripheral conductive lines CL connected to the peripheral contacts CT. The peripheral contacts CT and the peripheral conductive lines CL may include or be formed of a conductive material. The peripheral transistors PTR may be electrically connected to the plurality of first conductive lines **110**, the plurality of gate structures GS and the plurality of second conductive lines **150** through the peripheral contacts CT and the peripheral conductive lines CL.
(71) FIG. **28** is a cross-sectional view corresponding to the lines A-A′ and B-B′ of FIG. **2** to illustrate a semiconductor device according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the embodiments described with reference to FIGS. **1** to **5** will be mainly described for the purpose of ease and convenience in explanation.
(72) Referring to FIG. **28**, a plurality of first gate structures GS**1** may be disposed on the plurality of first conductive lines **110** and may intersect the plurality of first conductive lines **110**. The plurality of first gate structures GS**1** may be spaced apart from each other in the first direction D**1** and may extend in the second direction D**2**. Each of the plurality of first gate structures GS**1** may penetrate corresponding vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC. Each of the plurality of first gate structures GS**1** may penetrate the channel regions **120** of the corresponding vertical semiconductor patterns VC.
(73) A plurality of second gate structures GS**2** may be disposed on the plurality of first gate structures GS**1** and may be spaced apart from the plurality of first gate structures GS**1** in the third direction D**3**. The plurality of second gate structures GS**2** may extend in parallel to the plurality of first gate structures GS**1**. The plurality of second gate structures GS**2** may be spaced apart from each other in the first direction D**1** and may extend in the second direction D**2**. Each of the plurality of second gate structures GS**2** may penetrate corresponding vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC. Each of the plurality of second gate structures GS**2** may penetrate the channel regions **120** of the corresponding vertical semiconductor patterns VC. In addition, a plurality of third gate structures GS**3** may be disposed on the plurality of second gate structures GS**2** and may be spaced apart from the plurality of second gate structures GS**2** in the third direction D**3**. The plurality of third gate structures GS**3** may extend in parallel to the plurality of second gate structures GS**2**. The plurality of third gate structures GS**3** may be spaced apart from each other in the first direction D**1**

and may extend in the second direction D**2**. Each of the plurality of third gate structures GS**3** may penetrate corresponding vertical semiconductor patterns VC, spaced apart from each other in the second direction D**2**, of the plurality of vertical semiconductor patterns VC. Each of the plurality of third gate structures GS**3** may penetrate the channel regions **120** of the corresponding vertical semiconductor patterns VC.

(74) Each of the plurality of first gate structures GS**1**, the plurality of second gate structures GS**2** and the plurality of third gate structures GS**3** may include the gate electrode **130**, the ferroelectric pattern **132** and the gate insulating pattern **134**, described with reference to FIGS. **1** to **5**.

(75) The plurality of first gate structures GS**1**, the plurality of second gate structures GS**2** and the plurality of third gate structures GS**3** are illustrated as an example. However, the number of the gate structures stacked on the substrate **100** in the third direction D**3** is not limited thereto. Unlike FIG. **28**, a plurality of additional gate structures may be stacked on the plurality of third gate structures GS**3** and may be spaced apart from the plurality of third gate structures GS**3** in the third direction D**3**, or only two levels of gate structures may be included (e.g., GS**1** and GS**2**).

(76) Each of the plurality of vertical semiconductor patterns VC may surround an outer surface GS**1_5** of a corresponding first gate structure GS**1** of the plurality of first gate structures GS**1**, an outer surface GS**2**_S of a corresponding second gate structure GS**2** of the plurality of second gate structures GS**2**, and an outer surface GS**3**_S of a corresponding third gate structure GS**3** of the plurality of third gate structures GS**3**. The channel region **120** of each of the plurality of vertical semiconductor patterns VC may surround the outer surface GS**1**_S of the corresponding first gate structure GS**1**, the outer surface GS**2**_S of the corresponding second gate structure GS**2**, and the outer surface GS**3**_S of the corresponding third gate structure GS**3**. Each of the plurality of vertical semiconductor patterns VC and the corresponding first to third gate structures GS**1**, GS**2** and GS**3** penetrating therethrough may constitute a ferroelectric field-effect transistor having a multi-channel structure.

(77) According to some embodiments, support patterns **160** may be disposed on the substrate **100** and may support the plurality of first gate structures GS**1**, the plurality of second gate structures GS**2**, and the plurality of third gate structures GS**3**. One of the support patterns **160** may be in contact with end portions of the plurality of first gate structures GS**1**, end portions of the plurality of second gate structures GS**2**, and end portions of the plurality of third gate structures GS**3**. Another of the support patterns **160** may be in contact with other end portions of the plurality of first gate structures GS**1**, other end portions of the plurality of second gate structures GS**2**, and other end portions of the plurality of third gate structures GS**3**. Each of the support patterns **160** may include insulating patterns **168** and sacrificial patterns **164**, which are alternately stacked in the third direction D**3** on the substrate **100**. The sacrificial patterns **164** of the one of the support patterns **160** may be in contact with the end portions of the plurality of first gate structures GS**1**, the end portions of the plurality of second gate structures GS**2**, and the end portions of the plurality of third gate structures GS**3**. The sacrificial patterns **164** of the other of the support patterns **160** may be in contact with the other end portions of the plurality of first gate structures GS**1**, the other end portions of the plurality of second gate structures GS**2**, and the other end portions of the plurality of third gate structures GS**3**.

(78) According to the embodiments of the inventive concepts, the plurality of vertical semiconductor patterns may be two-dimensionally arranged on the substrate, from a plan view, and each of the plurality of gate structures may extend in one direction to penetrate corresponding ones of the plurality of vertical semiconductor patterns. Each of the plurality of gate structures may be disposed inside the corresponding vertical semiconductor patterns. Thus, it is possible to prevent disturbance between the gate structures adjacent to each other, and it may be easy to reduce a size of a memory cell structure including the plurality of gate structures and the plurality of vertical semiconductor patterns. As a result, a highly integrated semiconductor device and a method of manufacturing the same may be provided.

(79) In addition, each of the plurality of gate structures may include the gate electrode, the ferroelectric pattern surrounding the outer surface of the gate electrode, and the gate insulating pattern surrounding the outer surface of the ferroelectric pattern. Since the ferroelectric pattern surrounds the outer surface of the gate electrode, the intensity of the electric field applied to the ferroelectric pattern may be increased, and the intensity of the electric field applied to the gate insulating pattern may be reduced. Thus, the polarity properties of the ferroelectric pattern may be improved, and the endurance of the gate insulating pattern may be improved. As a result, a semiconductor device with improved operating characteristics and reliability and a method of manufacturing the same may be provided.
(80) While the inventive concepts have been described with reference to example embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the inventive concepts. Therefore, it should be understood that the above embodiments are not limiting, but illustrative. Thus, the scope of the invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing description.
(81) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “top,” “bottom,” and the like, may be used herein for ease of description to describe positional relationships, such as illustrated in the figures, for example. It will be understood that the spatially relative terms encompass different orientations of the device in addition to the orientation depicted in the figures.

Claims

1. A semiconductor device comprising: a plurality of first conductive lines extending in a first direction and spaced apart from each other in a second direction intersecting the first direction, the first direction and second direction being horizontal directions; a plurality of vertical semiconductor patterns disposed on the plurality of first conductive lines, respectively; a gate electrode crossing each of the plurality of first conductive lines and penetrating each of the plurality of vertical semiconductor patterns; a ferroelectric pattern between the gate electrode and each of the plurality of vertical semiconductor patterns; and a gate insulating pattern between the ferroelectric pattern and each of the plurality of vertical semiconductor patterns, wherein each of the plurality of vertical semiconductor patterns extends in a vertical direction perpendicular to the horizontal directions and includes a first dopant region and a second dopant region, which are spaced apart from each other in the vertical direction, and wherein the gate electrode, the ferroelectric pattern, and the gate insulating pattern are between the first dopant region and the second dopant region.
2. The semiconductor device of claim 1, wherein, where the gate electrode penetrates each of the plurality of vertical semiconductor patterns, the gate electrode is disposed inside of the respective vertical semiconductor pattern, and the respective vertical semiconductor pattern surrounds an outer surface of the gate electrode.
3. The semiconductor device of claim 2, wherein the ferroelectric pattern surrounds the outer surface of the gate electrode, and each of the plurality of vertical semiconductor patterns surrounds an outer surface of the ferroelectric pattern.
4. The semiconductor device of claim 3, wherein the gate insulating pattern surrounds the outer surface of the ferroelectric pattern, and each of the plurality of vertical semiconductor patterns surrounds an outer surface of the gate insulating pattern.
5. The semiconductor device of claim 2, wherein the outer surface of the gate electrode has a rounded shape when viewed in a cross-sectional view.
6. The semiconductor device of claim 2, wherein the outer surface of the gate electrode has an angular shape when viewed in a cross-sectional view.

7. The semiconductor device of claim 1, further comprising: a substrate, wherein each first conductive line of the plurality of first conductive lines is disposed on or in the substrate, and the first direction and the second direction are parallel to a top surface of the substrate, and wherein the vertical direction is perpendicular to the top surface of the substrate.
8. The semiconductor device of claim 7, wherein each of the plurality of vertical semiconductor patterns comprises: a channel region between the first dopant region and the second dopant region, and wherein the gate electrode, the ferroelectric pattern, and the gate insulating pattern penetrate the channel region of each of the plurality of vertical semiconductor patterns.
9. The semiconductor device of claim 7, wherein each first conductive line of the plurality of first conductive lines is buried in the substrate.
10. The semiconductor device of claim 7, further comprising: an additional gate electrode spaced apart from the gate electrode in the third vertical direction, the additional gate electrode extending in parallel to the gate electrode to penetrate each of the plurality of vertical semiconductor patterns; an additional ferroelectric pattern between the additional gate electrode and each of the plurality of vertical semiconductor patterns; and an additional gate insulating pattern between the additional ferroelectric pattern and each of the plurality of vertical semiconductor patterns.
11. The semiconductor device of claim 1, further comprising: a plurality of second conductive lines disposed on the plurality of vertical semiconductor patterns, wherein the plurality of second conductive lines are connected to the plurality of vertical semiconductor patterns, respectively.
12. The semiconductor device of claim 1, further comprising: a common conductive line disposed on the plurality of vertical semiconductor patterns, wherein the common conductive line is connected to the plurality of vertical semiconductor patterns.
13. The semiconductor device of claim 1, further comprising: a metal pattern between the ferroelectric pattern and the gate insulating pattern.
14. The semiconductor device of claim 1, further comprising: a substrate; a peripheral circuit structure on the substrate; and a lower interlayer insulating layer disposed on the substrate and covering the peripheral circuit structure, wherein the plurality of first conductive lines are disposed on the lower interlayer insulating layer.
15. A semiconductor device comprising: a plurality of vertical semiconductor patterns on a substrate, the plurality of vertical semiconductor patterns spaced apart from each other in a first direction and a second direction which are parallel to a top surface of the substrate and intersect each other, and the plurality of vertical semiconductor patterns extending in a third direction perpendicular to the top surface of the substrate; and a plurality of gate structures spaced apart from each other in the first direction and extending in the second direction on the substrate, wherein each of the plurality of gate structures penetrates each vertical semiconductor pattern of corresponding vertical semiconductor patterns spaced apart from each other in the second direction, of the plurality of vertical semiconductor patterns, wherein each of the plurality of gate structures comprises: a gate electrode penetrating the corresponding vertical semiconductor patterns; a ferroelectric pattern between the gate electrode and each of the corresponding vertical semiconductor patterns; a gate insulating pattern between the ferroelectric pattern and each of the corresponding vertical semiconductor patterns; first conductive lines disposed below the vertical semiconductor patterns and connected to the vertical semiconductor patterns; and second conductive lines disposed above the vertical semiconductor patterns and connected to the vertical semiconductor patterns.
16. The semiconductor device of claim 15, wherein for each gate electrode, each of the corresponding vertical semiconductor patterns surrounds an outer surface of the gate electrode, and wherein for each gate electrode, the ferroelectric pattern and the gate insulating pattern are disposed between the outer surface of the gate electrode and each of the corresponding vertical semiconductor patterns.
17. The semiconductor device of claim 16, wherein for each gate electrode, the ferroelectric pattern

surrounds the outer surface of the gate electrode, and the gate insulating pattern surrounds an outer surface of the ferroelectric pattern.

18. The semiconductor device of claim 15, wherein each of the plurality of gate structures further comprises: a metal pattern between the ferroelectric pattern and the gate insulating pattern.

19. The semiconductor device of claim 15, wherein each of the plurality of vertical semiconductor patterns comprises: a first dopant region and a second dopant region, which are spaced apart from each other in the third direction; and a channel region between the first and second dopant regions, wherein for each gate electrode, the gate electrode penetrates the channel regions of the corresponding vertical semiconductor patterns.

20. The semiconductor device of claim 15, wherein: the first conductive lines are buried in the substrate; and the second conductive lines are on the plurality of vertical semiconductor patterns, the first conductive lines extend in the first direction and are spaced apart from each other in the second direction, and each of the first conductive lines is connected to corresponding vertical semiconductor patterns, spaced apart from each other in the first direction, of the plurality of vertical semiconductor patterns, and the second conductive lines extend in the first direction and are spaced apart from each other in the second direction, and each of the second conductive lines is connected to corresponding vertical semiconductor patterns spaced apart from each other in the first direction.

21. A semiconductor device comprising: a plurality of first conductive lines each continuously extending lengthwise in a first direction, the plurality of first conductive lines spaced apart from each other in a second direction intersecting the first direction, the first direction and second direction being horizontal directions; a plurality of vertical semiconductor patterns disposed on the plurality of first conductive lines, so that each first conductive line is disposed below two or more vertical semiconductor patterns of the plurality of vertical semiconductor patterns in a third direction that is a vertical direction perpendicular to the horizontal direction; a gate electrode crossing the plurality of first conductive lines and surrounded by each vertical semiconductor pattern of a group of vertical semiconductor patterns of the plurality of vertical semiconductor patterns; a ferroelectric pattern between the gate electrode and each vertical semiconductor pattern of the group of vertical semiconductor patterns; and a gate insulating pattern between the ferroelectric pattern and each vertical semiconductor pattern of the group of vertical semiconductor patterns.